Fibonacci and Catalan Numbers

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1 Must Knows

The following things are all well-known in the olympiad community. There should be no problem in citing them without proof on a competition. If you do not know these results, you should work out the proofs. Hints are provided to guide you.

1.1 Fibonacci Numbers

Define the Fibonacci numbers by $F_0 = 0$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 1$.

Proposition 1.1 (Binet)
For all n,

$$F_n = \frac{(1+\sqrt{5})^n - (1-\sqrt{5})^n}{2^n\sqrt{5}}.$$

This can be proven easily by straightforward induction. However, this does not provide insight in how one would find the formula independently. For that purpose, we provide two outlines of proofs that enable you to solve general linear recurrences.

Proof. The first proof uses the characteristic polynomial of a recurrence relation. This method can be extended to solve most linear recurrences.

(a) Call a sequence (a_n) *Fibonacci-like* if it satisfies the recurrence $a_{n+1} = a_n + a_{n-1}$. Let λ be a real number. Show that if two sequences (a_n) and (b_n) are Fibonacci-like, then the sequences (λa_n) and $(a_n + b_n)$ are also Fibonacci-like.

(b) Let α and β be the roots of the polynomial $x^2 - x - 1 = 0$. Show that the sequences (α^n) and (β^n) are Fibonacci-like.

(c) Use the initial values of F_n to find real numbers λ_1 and λ_2 such that

$$F_n = \lambda_1\alpha^n + \lambda_2\beta^n.$$

(d) Conclude.

□

Proof. This proof uses generating functions, which can be used to solve a wide variety of problems, including more complex recursions.

(a) Let $F(X) = \sum_{n\geq 0} F_n X^n$ and solve for $F(X)$.

(b) Let α and β be the roots of $x^2 - x - 1$. Find constants A and B such that

$$F(X) = \frac{A}{1-\alpha X} + \frac{B}{1-\beta X}.$$

(c) Show that $\frac{1}{1-rX} = \sum_{n\geq 0} r^n X^n$.

(d) Conclude.

□

Proposition 1.2 (Periodicity)
The Fibonacci sequence is periodic modulo m for any positive integer m.

(a) Show that if there are two pairs (F_a, F_{a+1}) and (F_b, F_{b+1}) which are congruent mod m, then the sequence is eventually periodic. Use this to prove that (F_n) is eventually periodic mod m.

(b) Use the recurrence relation $a_{n-1} = a_{n+1} - a_n$ to show that the period starts at F_0.

Proposition 1.3 (Zeckendorf)
Every positive integer can be represented uniquely as the sum of one or more distinct Fibonacci numbers in a sum that does not include two consecutive Fibonacci numbers.

(a) Prove the existence of a Zeckdendorf representation using induction.

(b) Suppose that $F_m \leq n < F_{m+1}$. Prove that the Zeckendorf representation must include F_m. Use this to conclude uniqueness.

1.2 Catalan Numbers

The Catlan numbers are defined $C_0 = 1$ and $C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i}$ for $n \geq 0$.

Proposition 1.4 (Common appearances of Catalan numbers)
Each of the following are equal to C_n.

(a) The number of paths from a corner to the opposite corner along the grid lines of an $n \times n$ grid which use exactly $2n$ moves and do not cross the diagonal of the square.

(b) The number of expressions containing n pairs of parentheses which are correctly matched.

(c) The number of ways of draw n line segments among $2n$ points on a circle such that no two segments intersect (even at an endpoint).

(d) The number of triangulations of a convex polygon with $n + 2$ vertices.

(e) The number of full binary trees with $n + 1$ leaves. (A full binary tree is a graph that is a tree where every node has either 0 or 2 children.)

(f) The number of ways to place the numbers $1, 2, \ldots, 2n$ in a $2 \times n$ grid such that each row and column is increasing.

The proof is left as an exercise.

Proposition 1.5
For all n,

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

Proof. (a) Label the grid such that the top right corner is (n, n). Find a point (a, b) such that there is a bijection between the number of paths to (a, b) and paths that cross the diagonal.

(b) Conclude a formula for C_n.

□

Proof. (a) Let $C(X) = \sum_{n\geq 0} C_n X^n$. Show that

$$C(X) = \frac{1-\sqrt{1-4X}}{X}.$$

(b) Apply the binomial theorem to $\sqrt{1-4X}$ to solve for C_n.

□

2 Good to Knows

The following are properties of Fibonacci numbers that occasionally show up in problems. These can be useful to know and can most likely be cited without proof. However, you most likely will not remember all of these results and may need to refind them on your own during a contest.

(a) For all n,

$$F_1 + F_2 + \cdots + F_n = F_{n+2} - 1.$$

(b) For all n.

$$F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}.$$

(c) For all n and r,

$$F_n^2 - F_{n+r}F_{n-r} = (-1)^{n-r}F_r^2.$$

(d) For all $m > n$,

$$F_m F_{n+1} - F_{m+1}F_n = (-1)^n F_{m-n}.$$

(e) If $m \mid n$, then $F_m \mid F_n$.

(f) For all m and n,

$$\gcd(F_m, F_n) = F_{\gcd(m,n)}.$$

(g) Define the *Lucas numbers* L_n by $L_1 = 1$, $L_2 = 3$ and $L_{n+1} = L_n + L_{n-1}$ for $n \geq 1$. Show that $F_{n-1} + F_{n+1} = L_n$.

3 Problems

3.1 Combinatorics

Problem 3.1 (Bangladesh 2020)**.** Urmi is playing a game on a computer. If the computer screen displays the number x, then in the next move, Urmi can do one of the following:

1. Replace x by $4x + 1$.
2. Replace x by the largest integer not greater than $x/2$.

Initially the computer screen displays 0. How many different integers less than or equal to 2020 can Urmi achieve through a sequence of moves? It is permitted for the number displayed on the screen to exceed 2020 during the sequence.

Problem 3.2 (USA TSTST 2019)**.** Let $\mathcal{S}$ be a set of 16 points in the plane, no three collinear. Let $\chi(\mathcal{S})$ denote the number of ways to draw 8 lines with endpoints in $\mathcal{S}$, such that no two drawn segments intersect, even at endpoints. Find the smallest possible value of $\chi(\mathcal{S})$ across all such $\mathcal{S}$.

Problem 3.3 (RMM 2018)**.** Let n be positive integer and fix $2n$ distinct points on a circle. Determine the number of ways to connect the points with n arrows (oriented line segments) such that all of the following conditions hold: each of the $2n$ points is a startpoint or endpoint of an arrow; no two arrows intersect; and there are no two arrows $\overrightarrow{AB}$ and $\overrightarrow{CD}$ such that A, B, C and D appear in clockwise order around the circle (not necessarily consecutively).

Problem 3.4 (China 2020)**.** Consider a triangulation of a convex polygon with 20 vertices. A subset of 10 of the 37 line segments in a triangulation is called a *perfect matching* if covers all 20 vertices. Among all possible triangulations, find the maximum number of perfect matchings.

Problem 3.5 (Putnam 2020)**.** Let a_n be the number of sets S of positive integers for which

$$\sum_{k\in S} F_k = n.$$

Find the largest number n such that $a_n = 2020$.

Problem 3.6 (Indonesia TST 2020)**.** Given a natural number n, define B_n as the set of all sequences $b_1, b_2, \ldots, b_n$ of length n such that $b_1 = 1$ and for every $i = 1, 2, \ldots, n$, then we have

$$b_{i+1} - b_i \in \{1, -1, -3, -5, -7, \ldots\}$$

Where $b_i > 0$ for all i. Find a closed form of $|B_n|$

Problem 3.7 (Estonia, USA)**.** A sequence of positive integers $(a_n)_{n\geq 1}$ is of Fibonacci type if it satisfies the recursive relation $a_{n+2} = a_{n+1} + a_n$ for all $n \geq 1$. Is it possible to partition the set of positive integers into an infinite number of Fibonacci type sequences?

3.2 Number Theory

Problem 3.8. Prove the properties listed in Section 2.

Problem 3.9. Let k be an integer such that $p = 5k + 1$ is a prime. Show that the period of the Fibonacci sequence modulo p divides $p - 1$.

Problem 3.10 (ELMO 2020)**.** Let k be a positive integer. Suppose that for every positive integer m there exists a positive integer n such that $m \mid F_n - k$. Must k be a Fibonacci number?

Problem 3.11 (Harvard-MIT Mathematics Tournament 2017)**.** What is the period of the Fibonacci sequence modulo 127?

Problem 3.12 (USA TSTST 2019)**.** Suppose P is a polynomial with integer coefficients such that for every positive integer n, the sum of the decimal digits of $|P(n)|$ is not a Fibonacci number. Must P be constant?

Solutions

3.1 Consider the binary representation of x. Move 1 is equivalent to appending "01" to the end of x, and Move 2 is equivalent to removing the final digit of x. Therefore it is impossible for the sub-string "11" to ever appear in the binary representation of x. It is not hard to see that any other integer is obtainable.

Because any number between 2021 and 2048 contains the sub-string "11", it is sufficient to count the number of binary strings of length 11 which do contain "11." Call such a string *spacy.* In general, let a_n denote the number of strings of length n such that this is true. By inspection, $a_1 = 2$ and $a_2 = 3$.

For any other n, we distinguish into two cases. If the string ends with a 0, then the first $n-1$ characters must be a spacy string. There are a_{n-1} ways to do this. If the string ends with a 1, its immediate predecessor must be a 0. The remaining $n-2$ characters must also form a spacy substring. There are a_{n-2} ways for this to occur.

By induction, $a_n = F_{n+2}$, so $a_{11} = F_{13} = 233$

3.2 Replace 16 by $2n$. The answer is C_n, the n-th Catalan number. Proceed by induction on n. The base case $n = 1$ is simple.

Consider a point P on the convex hull of $\mathcal{S}$ and enumerate the remaining points $A_1, A_2, \cdots, A_{2n-1}$ such that the points lie in clockwise order with respect to P. Construct the line ℓ from P to A_{2k+1}. Then there are precisely $2k$ points to the right of ℓ and $2n-2k-2$ points to the left of ℓ. Call these sets of points A and B, respectively. No segment drawn among the points in A can intersect a segment drawn among the points in B; hence

$$\chi(S) \geq \sum_{k=0}^{n-1} \chi(A_k)\chi(B_k) \geq \sum_{k=0}^{n-1} C_k C_{n-1-k} = C_n.$$

It is well known that any convex polygon achieves equality. Thus the answer is $C_9 = 1430$.

3.3 The answer is $\binom{2n}{n}$. Is it well known that there are $\frac{1}{n+1}\binom{2n}{n}$ ways to connect the points without orientation such that none of the segments intersect. We shall show that for every such configuration, there are exactly n ways to orient the arrows which fulfill the final condition.

We proceed by induction on n. The base case $n = 1$ is clear.

It is clear that there must exist an edge which connects consecutive points on the circle. If this edge is counter-clockwise, then we may disregard this edge and orient the remaining $2n-2$ arrows in n distinct ways. Otherwise, all other arrows must point counter-clockwise. This gives a total of $n+1$ ways.

3.4 Replace 20 by $2n$. Then the answer is F_{n+1}. Proceed by induction on n. The base cases $n = 2$ and $n = 3$ are checked by hand.

Label the polygon $P_1, P_2, \ldots, P_{2n}$. Call a diagonal P_iP_j *odd* if $i-j$ is even and *even* otherwise. It is clear that any perfect matching must contain only even edges.

Let e denote the even diagonal which minimises $|i-j|$. (If no such diagonal exists, then there are simply 2 perfect matchings.) WLOG, assume that this diagonal is $P_{2n}P_{2k+1}$ for some integer k. By the minimality of e, each point $P_1, \ldots P_{2k}$ must be connected to one of its immediate neighbours. In particular, P_1 is connected to either $P_{2n} or P_2$. Either way forces the choice of edges for each of $P_2, P_3, \ldots, P_{2k}$. If P_{2n} is selected then the points $P_{2k+2}, P_{2k+3}, \ldots P_{2n-1}$ need to be perfectly matched matched among themselves, for which there are at most F_{n-k-1} possible ways.

Likewise if P_1 is connected to P_2, then that $P_{2k+1}, P_{2k+2}, \ldots, P_{2n}$ need to be perfect matched, which happen in at most F_{n-k} ways. Hence, there are at most

$$F_{n-k-1} + F_{n-k} \leq F_{n-2} + F_{n-1} \leq F_n$$

perfect matchings.

The construction is left as an exercise to the reader.

3.5 Replace 2020 by n. We claim that the answer is $F_{2n} - 1$.

Claim. For all $k > 1$, $a_{F_{2k}} > k$.

Proof. Proceed by induction on k. The base case $n = 2$ is checked to work. Then

$$F_{2k} = F_{2k-1} + F_{2k-2}.$$

There are at least k representations of F_{2k-2} none of which contain F_{2k-1}. In addition to $S = \{2n\}$, this gives $k + 1$ representations. □

Claim. Let $a > 1$ be an integer. Then there are at least $k - a$ representations of F_{2k-2} using only the Fibonacci numbers which are at least F_{2a}.

Proof. The proof is also by induction on k. The base case $k = a + 1$ is clear. Then $F_{2k} = F_{2k-1} + F_{2k-2}$. Since F_{2k-2} has at least $k - a$ representations, it follows that F_{2k} hast at least $k - a + 1$ representations, completing the induction. □

Claim. Let m be an integer such that $m > F_{2k-2}$. Then $a_m \geq k$. Write $m = F_{2k-2} + r$ and let a denote the greatest integer such that $F_{2a} < r$. By the first claim, there are at least $a + 1$ representations of F_{2a}. By the second claim, we can write F_{2k-2} in at least $k - a$ ways which do not overlap with any of the representations of F_{2a}. Therefore

$$a_m \geq (a + 1)(k - a) \geq k \iff k \geq a + 1,$$

which is obviously true.

Claim. For all n, $a_{F_{2k}-1} = k$.

Proof. Once again, we use induction on k. Since

$$F_1 + F_2 + \cdots + F_{2k-2} = F_{2k} - 1,$$

then there is only one possible set S for which $2k - 1 \notin S$. On the other hand, there are $a_{F_{2k-2}-1}$ for which $2k - 1 \in S$. Summing these cases completes the induction. □

Combining all these claims proves the result.

3.6 Consider the transformation. $a_i = \frac{b_i - i}{2}$. Then the problem is equivalent to finding the number of non-decreasing sequences (a_i) of of positive integers with length n such that $a_i \leq \frac{i-1}{2}$ for each i. Equivalently, this is the number of monotonic paths from the bottom-left to top-right corner of a $n \times \lceil n/2 \rceil$ rectangular grid that does not cross the main diagonal. This inspires us to use our Catalan toolbox. It is necessary to break into cases based on the parity of n. If $n = 2m$, then the grid is $2m \times m$. If $n = 2m + 1$, then the grid is $(2m + 1) \times (m + 1)$.

Case. $n = 2m$. We shall invoke a more general result for a $an \times n$ grid. Before that, however, we shall detail another proof for the explicit formula of the Catalan numbers that we not cover in class.

Consider the representation of the Catalan numbers corresponding to monotonic paths on an $n \times n$ grid. For a path P, define the *exceedance* of P by the number of *horizontal* edges which are above the diagonal. We claim that the number of paths which have an exceedance of k is independent of k.

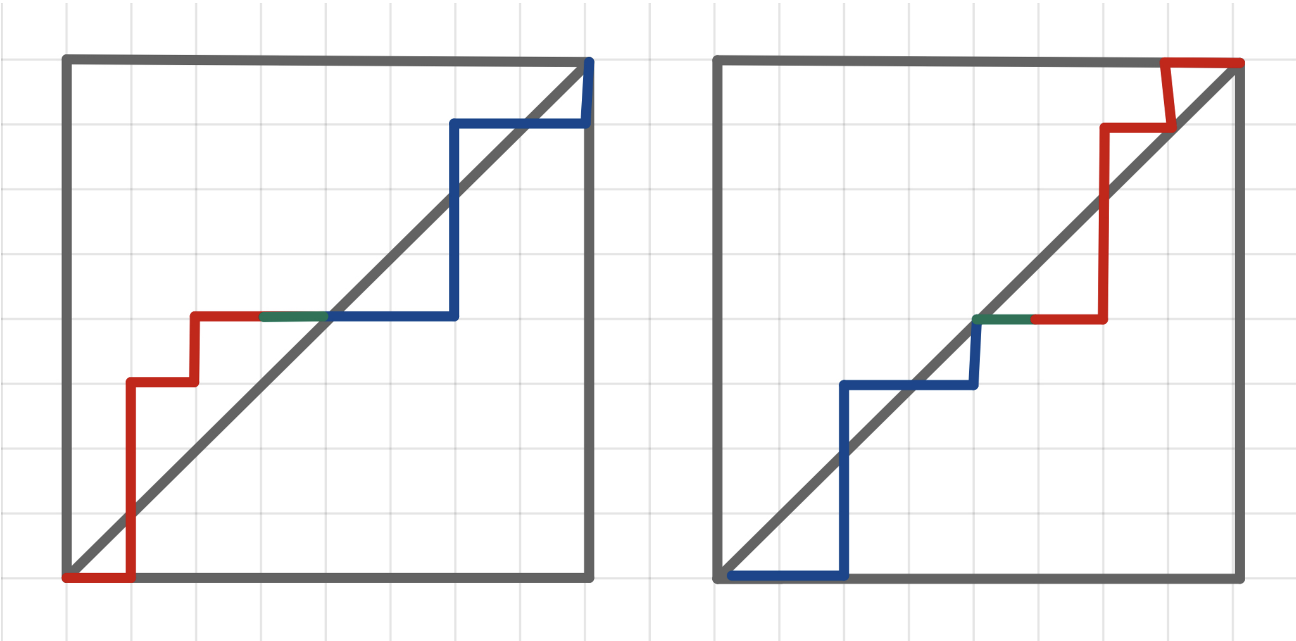
Consider the following procedure for any path which crosses the diagonal:

(a) Mark the spot where P first crosses *back* over to the lower half of the diagonal X.

(b) Label the edge immediately before X by e. Note that e is horizontal.

(c) Denote the subpath that precedes e by A and the subpath that comes after e by B.

(d) Create a new path by chaining B, e, then A.



It is not hard to verify that this decreases the exceedance of P by 1. Furthermore, this process is reversible. In the reverse process, the edge e is the last horizontal edge that originates on the diagonal.

This establishes a bijection between paths of exceedance j and k, so there are $\frac{1}{n+1}\binom{2n}{n}$ paths of any exceedance.

The same idea can be used to solve the $an \times n$ problem. Instead, use a $an \times an$ grid where one can only move a moves at a time. Then the exact same procedure can be used to increase or decrease the exceedance of a path by 1. Consequently, the number of paths which stay below the diagonal is equal to

$$\frac{1}{an+1}\binom{an+n}{n}.$$

In our case, this gives an answer of

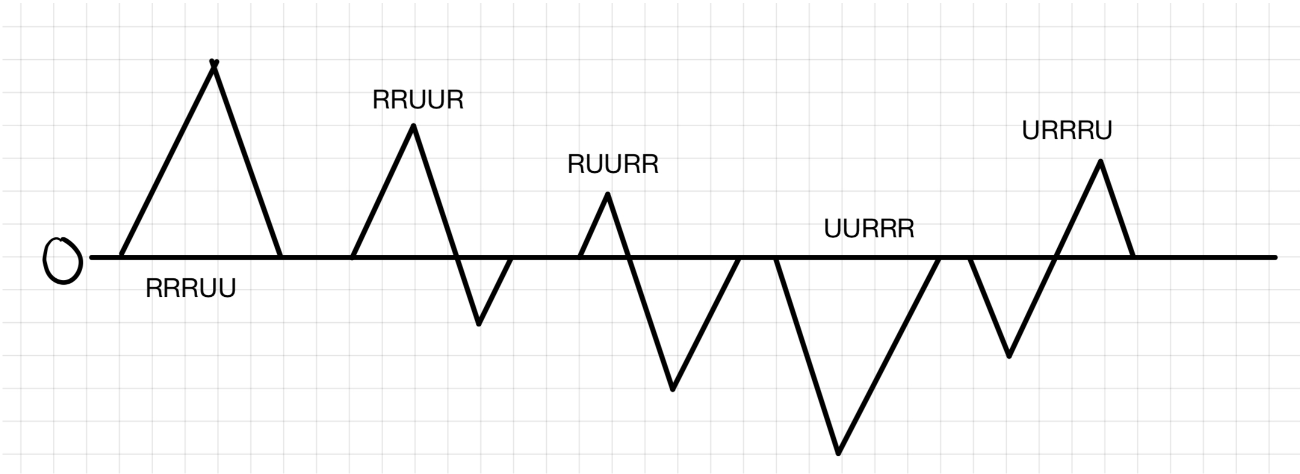
$$\frac{1}{2m+1}\binom{3m}{m}.$$

Case. $n = 2m+1$. We also solve a more general problem here, which is the number of paths which stay below the diagonal on an $a \times b$ grid where a and b are coprime. The solution here is rather tricky.

Consider an arbitrary path P on the $a \times b$ grid and consider all cyclic permutations of P. We claim that exactly one of these cyclic permutations passes under the diagonal.

Consider a random walk on $\mathbb{R}$. Starting at zero, every move right corresponding to an increase of b and every move up corresponds to a decrease of a. Then a path P lies beneath the diagonal if and only if every value of the walk is non-negative.

Clearly the path both starts and ends at 0. Every value on the walk (excluding the starting value) must be unique, or else there exists integers x and y such that $ax + by = 0$ and $x + y < a + b$, which is impossible. In particular, there exists a unique minimum to the walk. Now consider the cyclic shift such that the minimum of the walk occurs at the final step. This implies that 0 is the minimum of the walk, so the path lies beneath the diagonal. In contrast, every other cyclic shift has the minimum occur somewhere other than the end. Since the minimum value is unique, this implies that the walk dips below zero, so the path goes above the diagonal.



(A picture is included for reference for a 3×2 grid and the path RRRUU.)

Therefore precisely $\frac{1}{a+b}$ of the total number of paths lie under the diagonal. For our specific case, that gives an answer of

$$\frac{1}{3m+2}\binom{3m+2}{m+1}.$$

Our final answer is

$$\begin{cases} \frac{1}{2m+1}\binom{3m}{m} & \text{if } n = 2m, \\ \frac{1}{3m+2}\binom{3m+2}{m+1} & \text{if } n = 2m+1. \end{cases}$$

3.7 Solution 1. Consider the Zeckendorf representations of all positive integers. We partition $\mathbb{N}$ into classes by the following equivalence relation: two numbers are equivalent if they're identical up to trailing zeroes. It is clear that each class forms a Fibonacci-like sequence so we are done.

Solution 2. Given a natural number x, define a sequence (a_n) by $a_1 = x$ and

$$a_{n+1} = \left\lfloor \phi a_n + \frac{1}{2} \right\rfloor,$$

where $\phi = \frac{1+\sqrt{5}}{2}$ We claim that (a_n) is Fibonacci-type. It suffices to show that

$$\left\lfloor \phi \left\lfloor \phi x + \frac{1}{2} \right\rfloor + \frac{1}{2} \right\rfloor = \left\lfloor \phi x + \frac{1}{2} \right\rfloor + x$$

for any integer x. Notice that

$$\begin{aligned} \left\lfloor \phi \left\lfloor \phi x + \frac{1}{2} \right\rfloor + \frac{1}{2} \right\rfloor &= \left\lfloor \left(1 + \frac{1}{\phi}\right) \left\lfloor \phi x + \frac{1}{2} \right\rfloor + \frac{1}{2} \right\rfloor \\ &= \left\lfloor \phi x + \frac{1}{2} \right\rfloor + \left\lfloor \frac{1}{\phi} \left\lfloor \phi x + \frac{1}{2} \right\rfloor + \frac{1}{2} \right\rfloor \end{aligned}$$

Since $\left| \left\lfloor \phi x + \frac{1}{2} \right\rfloor - \phi x \right| < \frac{1}{2}$, it follows that $\phi^{-1} \left\lfloor \phi x + \frac{1}{2} \right\rfloor$ is within $\frac{\phi^{-1}}{2} < \frac{1}{2}$ of x.

Thus we partition $\mathbb{N}$ in the following way. Construct a Fibonacci type sequence with $a_1 = 1$ in the previously described way. Then at every next step, take the smallest number x not appearing in any of the used Fibonacci type sequences and make a new sequence in the aforementioned way with $a_1 = x$.

Obviously this process includes every natural number in some sequence. Moreover, because the function $x \to \left\lfloor \phi x + \frac{1}{2} \right\rfloor$ is injective, this partitions $\mathbb{N}$ without overlap.

3.8 Proofs are easily available online.

3.9 By the quadratic reciprocity law, there exists an integer z such that $z^2 \equiv 5 \pmod{p}$. Then by Binet's formula,

$$F_n \equiv \frac{(1+z)^n - (1-z)^n}{2^n z}.$$

Using Fermat,

$$F_{p-1} \equiv \frac{1}{z} \cdot \frac{1-1}{1} \equiv 0 \pmod p$$
$$F_p \equiv \frac{1}{z} \cdot \frac{(1+z)-(1-z)}{2} \equiv 1 \pmod p.$$

3.10 There are many approaches. See link.
https://artofproblemsolving.com/community/c6h2210477p16724051

3.11 Solution 1. Let $p = 127$. The answer is 256. By Binet's formula,

$$F_n = \frac{1}{2^{n-1}} \sum_{j=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{n}{2j+1} 5^j.$$

Thus

$$F_{p+1} \equiv \frac{1}{2^p}\left(\binom{p+1}{1}5^0 + \binom{p+1}{p}5^{\frac{p-1}{2}}\right) \equiv \frac{1+5^{\frac{p-1}{2}}}{2} \pmod p.$$

By the law of quadratic reciprocity,

$$5^{\frac{p-1}{2}} \equiv \left(\frac{5}{p}\right) \equiv \left(\frac{p}{5}\right) \equiv -1 \pmod p.$$

Therefore $p \mid F_{128} \mid F_{256}$. Additionally,

$$F_{2p} \equiv \frac{1}{2^{2p-1}}\binom{2p}{p}5^{\frac{p-1}{2}} \equiv \frac{1}{2} \cdot 2 \cdot -1 \equiv -1 \pmod p.$$

Thus

$$F_{2p+2} \equiv F_0 \pmod p$$
$$F_{2p} \equiv F_{-2} \pmod p$$

so the period divides $2p + 2 = 256$.

It suffices to check that the period is not 128. We check

$$F_p \equiv \frac{1}{2^{p-1}}\binom{p}{p}5^{p-1}2 \equiv -1 \pmod p.$$

Hence $F_p \not\equiv F_{-1} \pmod p$, so the period does not divide 128.

Solution 2. Let $p = 127$. We may wish to this problem in a similar manner to 3.9 using Binet's formula, but that does not work because 5 is not a quadratic residue modulo p. Fortune favors the bold, however, so decide to define an "imaginary" quantity $\sqrt{5} \pmod p$. In other words, consider

$$\mathbb{F}_p[\sqrt{5}] = \{a + b\sqrt{5} \mid a, b \in \mathbb{F}_p\}.$$

This ring (with addition and multiplication defined in the expected manner) is a field. (The proof is left as an exercise.)

Consider the function $F : x \to x^p$. (This function is known as the Frobenius endomorphism.) Prove each of the following properties:

1. An element x is a fixed point of F if and only if $x \in \mathbb{F}_p$. (i.e. x has no $\sqrt{5}$ part.
2. For any two elements $x, y \in \mathbb{F}_p[\sqrt{5}]$,

$$F(x+y) = F(x) + F(y) \text{ and } F(xy) = F(x)F(y).$$

3. For any element x and polynomial P, $F(P(x)) = P(F(x))$.

Consider $P(X) = X^2 - X - 1$, which has the roots of α and β. Since α is not a fixed point of F,

$$P(F(\alpha)) = F(P(\alpha)) = 0 \implies F(\alpha) = \beta.$$

Likewise $F(\beta) = \alpha$.

Returning to the problem at hand, we may employ Binet's formula to obtain

$$\begin{aligned} F_{2p+2} &= \frac{\alpha^2\beta^2 - \beta^2\alpha^2}{\sqrt{5}} = 0 \\ F_{2p+3} &= \frac{\alpha^3\beta^2 - \beta^3\alpha^2}{\sqrt{5}} = (\alpha\beta)^2 = 1. \end{aligned}$$

Therefore the period divides $2p + 2 = 256$. On the other hand,

$$F_{p+2} = \alpha\beta = -1,$$

so the period does not divide 128. Hence the answer is 256.

Remark. One may be hesitant to accept this solution, objecting that this supposed $\sqrt{5} \pmod p$ does not actually exist. In doing so one would be following in the footsteps of great mathematicians like Rene Descartes, who objected to the paradigm of complex numbers because the quantity i was completely "imaginary" (thereby creating the moniker "imaginary numbers"). However, as mathematicians like Euler and Gauss would soon show, this position was quite incorrect. It matters whether or not the number "actually exist" regardless of what that vague phrase even means. What is important is whether or not the paradigm is sound and logical. If one wishes to avoid the notion of an "imagined" number, one may simply construct a unit i, such that

$$\begin{aligned} (a + bi) + (c + di) &= (a + c) + (b + d)i \\ (a + bi)(c + di) &= (ab + 5cd) + (ad + bc)i, \end{aligned}$$

with $a, b, c, d \in \mathbb{F}_p$. This, for all intents and purposes, behaves just like $\sqrt{5}$, allowing us to apply Binet's formula in the manner above.

3.12 Assume without loss of generality that the leading coefficient of P is positive, and let $S(N)$ denote the sum of the digits of N. Write out $P(n) = a_0 + a_1x + \cdots + a_nx^n$. The main idea is to plug in $n = 10^e$ in order to wrest some level of control over the output of $S \circ P$. Naively plugging $n = 10^e$, we obtain that, for large enough e,

$$S(P(10^e)) = 9me + C,$$

where m denotes the number of coefficients of P which are negative.

If the Fibonacci numbers were surjective (mod $9m$), this would finish the problem. Unfortunately, this is not the case. However, one may notice that the Fibonacci numbers are surjective (mod 9); thus we we are motivated to try to prove the following crucial claim.

Claim. Call a polynomial *good* if it all coefficients are positive except for the second greatest coefficient, which is strictly negative. Given a polynomial $P \in \mathbb{Z}[X]$ there exists another polynomial $Q \in \mathbb{Z}[X]$ such that $P(Q(x))$ is good.

Proof. Write

$$P(Q(x)) = a_nQ(x)^n + a_{n-1}Q(x)^{n-1} + \cdots + a_0.$$

Notice that by multiplying Q by a suitably large constant, it suffices to find a Q such that $Q(x)^n$ is good.

Let $Q(x) = x^3 - \varepsilon x^2 + x + 1$. Then

$$Q(x)^n = x^{3n} - \varepsilon x^{3n-1} + \sum_{k=0}^{3n-2} x^k c_k,$$

where there exists some constant C such that

$$c_k < 1 - C\varepsilon.$$

Thus by taking ε sufficiently small, we can guarantee that $Q(x)^n$ is good. Then by multiplying Q by a suitable constant D such that $D\varepsilon \in \mathbb{Z}$. $\square$

Then for large enough e,

$$S(P(Q(10^e))) = 9e + C$$

for some constant C. Since the Fibonacci numbers are surjective and periodic modulo 9, we are done.